LLM	Instruction Retention	Inference Memory	Reliable Version Editing	Self-Coherence	Average
GPT-4o (August 2024)	14.29	5.08	17.07	13.64	12.52
Llama 3.1 405B Instruct	12.86	16.95	4.88	25.0	14.92
Mistral Large	21.43	9.32	7.32	20.45	14.63
Claude 3.5 Sonnet (June 2024)	**58.57**	37.29	24.39	**45.45**	**41.42**
Gemini 1.5 Pro (August 27 2024)	31.43	15.25	19.51	13.64	19.96
o1-preview	34.29	**41.53**	**39.02**	34.09	37.23

Table 2: Human-evaluated performance of 6 frontier LLMs on MultiChallenge.

LLM	Instruction Retention	Inference Memory	Reliable Version Editing	Self-Coherence	Average
GPT-4o (August 2024)	12.86	5.93	14.63	18.18	12.9
Llama 3.1 405B Instruct	14.29	16.95	9.76	25.0	16.5
Mistral Large	18.57	6.78	9.76	18.18	13.32
Claude 3.5 Sonnet (June 2024)	**61.43**	37.29	26.83	**45.45**	**42.75**
Gemini 1.5 Pro (August 27 2024)	31.43	16.1	19.51	18.18	21.3
o1-preview	28.57	**38.98**	**39.02**	36.36	35.73

Table 3: Automatic evaluated performance of 6 frontier LLMs on MultiChallenge. We adopt auto-eval with instance-level rubrics illustrated in Section 3.2 for automatic evaluation of MultiChallenge.

Challenge Category	Auto-Eval Baseline(%)	Auto-Eval with IR (Ours)(%)
Instruction Retention	44.44	**92.26**
Inference Memory	37.53	**94.62**
Reliable Version Editing	31.82	**94.85**
Self-Coherence	31.05	**94.12**
Overall	37.33	**93.95**

Table 4: Alignment between Human Rater and Evaluation Methods across categories in MultiChallenge.

conducting multi-turn conversations with human users, reaching a 38.3% average accuracy score on MultiChallenge, significantly outperforming other LLMs. o1-preview follows with 34.6, also significantly outperforming the rest LLMs. Gemini 1.5 Pro (August 27 2024) achieves 19.1 and wins the third place.

When zooming into different challenge categories, Table 2 also shows that different LLMs have different expertise on the 4 challenges that we test in MultiChallenge. Although Claude achieves the best general performance, it falls behind on Reliable Versioned Editing and Information Memory compared to o1-preview. Although Gemini 1.5 Pro ranks behind o1-preview on average scores, Gemini 1.5 Pro 's performance is on par with o1-preview on instruction retention. The performance differences of frontier LLMs across these challenges validates the effectiveness of MultiChallenge's design in defining 4 categories that target distinct LLM capabilities.

In Table 3, we also use our automatic evaluation method to assess the 6 frontier LLMs on MultiChallenge. The auto-eval results are almost identical to human-eval results, and the trends and rankings also stay consistent between human-eval and auto-eval results. We adopt GPT-4o as the base model for the LLM judge, since it has very high alignment, as demonstrated in Table 4. We also try using Claude as the base model for the LLM judge and reach the same conclusions. We also present detailed alignment results of our auto-eval method with human raters in Section 5.2.

5.2 Analysis

The alignment between Automatic Evaluation and Human Evaluation. We calculate the alignment of our auto-eval with instance-level rubric questions with human raters and show them in Table 4. The alignment score is calculated with all data with human labels in Table 2. Results show 93.91% alignment score on average. The alignments on 4 individual challenges are all above 90%. We implemented a baseline auto-eval method for

LLM	Instruction Retention	Inference Memory	Reliable Version Editing	Self-Coherence	Average
Llama-3.2-3B-Instruct	15.94	8.85	**36.59**	6.0	16.85
Llama-3.3-70B-Instruct	**33.33**	15.04	24.39	**20.0**	**23.19**
Qwen2-72B-Instruct	27.54	7.96	26.83	20.0	20.58
Qwen2.5-14B-Instruct	15.94	**15.93**	24.39	12.0	17.07
Qwen2.5-72B-Instruct	21.74	17.70	12.20	16.0	16.91
Mixtral-8x7B-Instruct-v0.1	13.04	7.08	12.20	12.0	11.08
Mixtral-8x22B-Instruct-v0.1	15.94	3.54	15.94	8.00	14.18

Table 5: Automatic evaluated performance of open source LLMs on `MultiChallenge`.

comparison, in which we directly apply frontier LLMs as judge by providing the full multi-turn conversation history and prompting them with winning/losing criteria of the corresponding challenge category, to evaluate model responses. Results in Table 4 show that this method obtains 36.01% alignment score, significantly lower than our approach. Both GPT-4o and Claude are used as the base model of LLM judge of the two auto-eval methods in this experiment, and the same conclusion is reached.

Is the number of turns correlated to LLM performance? The number of turns vary for different test examples in `MultiChallenge`. We would like to see whether the number of turns directly correlates to the difficulty level of test examples. Figure 4 shows that for all models, the performance on the benchmark does not show any visible trend as the number of turns increases, indicating that the challenge of `MultiChallenge` is not caused simply by the length of the conversations. Although being a multi-turn benchmark, the lengths of test examples are still pretty small compared the current models' max token lengths (as illustrated in Table 1). Therefore adding turns (length) to the example doesn't add on extra difficulties. It is potentially the inherent reasoning required given the conversation context that makes `MultiChallenge` challenging, instead of lengths.

How do open source models perform on `MultiChallenge`? We also assess some most recent open source models on `MultiChallenge` with our auto-eval methods and present the results in Table 5. Since `MultiChallenge` only consists of test examples in which at least 3 of the 6 frontier models fail (details in Section 4.1), `MultiChallenge` can be potentially biased against these 6 frontier models in Table 3, compared to the open source

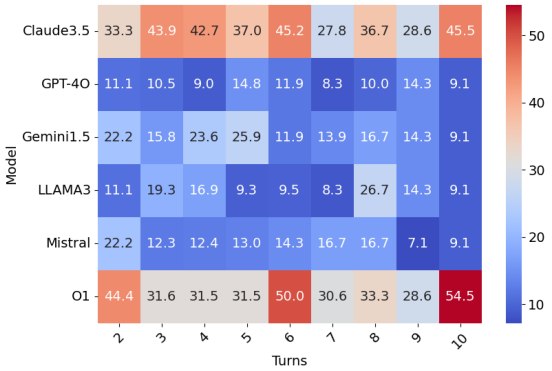

Figure 4: The LLM performance distribution across numbers of turns.

models in Table 5. However, given the potential bias against the frontier models, we still observe from Table 5 that all open source models fall behind top-performing closed source mooels such as Claude 3.5 Sonnet and o1-preview.

How do model size and release date relate to performance? We examine the relationship between model performance and two key factors: model size (in terms of parameters) and release date. Figure 5 illustrates the performance of various models across different parameter sizes. On average, we observe that larger models, particularly those in the 70-72B parameter range, tend to outperform smaller ones. Figure 8 in Appendix presents the performance of models over time. We note a general trend of improvement, with more recent models typically achieving higher scores. However, this progression is not strictly linear. For example, Claude 3.5 Sonnet, released in June 2024, outperforms several models released later.

How effective is the synthetic data generation? We measure the effectiveness of the synthetic data generated from `MMSE` by the Average Minutes per Test example (AMT) required for hu-

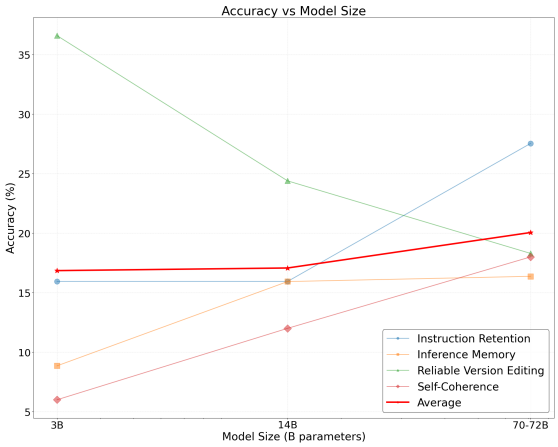

Figure 5: Open Source Model performance across different parameter sizes.

man editing. The more effective MMSE is, the less AMT of human editing would be required. On average, trained human raters require 73.6 minutes to accomplish one test example if they edit from synthetic data generated by MMSE. While without synthetic data, it takes trained human raters 154.4 minutes on average to finish producing one test example from scratch. This comparison shows that MMSE successfully reduced required AMT by half, therefore significantly reduced cost.

How much human editing was required? Although the results above show that synthetic data from MMSE significantly reduced human time required for creating each test examples, human reviewing and editing is still critical for keeping MultiChallenge natural and challenging. To assess the degree of modifications by human raters, we apply the fuzzywuzzy library[1] to calculate Levenshtein-distance-based string similarities of the original synthetic example vs the human-edited version. The average string similarity between all original synthetic examples and the final human-edited examples in MultiChallenge is 74.5%, indicating 25.5% of difference led by human editing.

Case study of LLM failures We provide examples of how the 6 frontier LLMs provide improper responses for all 4 challenges in Appendix A.8.

6 Conclusion

Our introduction of MultiChallenge marks a significant advancement in evaluating large language models' (LLMs) capability at conducting multi-turn conversations with human users, a crucial but previously overlooked aspect. Through a novel AI-Human collaborative construction, MultiChallenge offers a comprehensive benchmark covering 4 multi-turn challenges that are realistic and currently challenging to frontier LLMs. The LLM as judge with instance-level rubrics also makes auto-eval for this challenging benchmark possible and trustworthy.

Limitations

Our research marks a significant step forward in assessing the multi-turn conversation capabilities of large language models (LLMs). However, it is not without its challenges. One constraint is in order to make the LLM as judge trustworthy, we cannot include examples in which even the corresponding instance-level rubric question is beyond current frontier LLM capability. This potentially limits the difficulty level of the entire evaluation benchmark. We keep in private such difficult test examples and keep monitoring the evolvement of LLM capability. We will release the difficult data set to the public once their auto-eval using LLM is possible. Another constraint is this benchmark is inevitably biased against the 6 frontier models listed above, due to the fact that the test examples are picked according to the 6 frontier models' common failures. However, given the potential bias against the frontier models, we still observe that all open source models we tested later fall behind top-performing closed source models such as Claude 3.5 Sonnet and o1-preview. Moreover, the comparison among all the future released models of the 6 frontier LLM families and existing open source models would still be fair on MultiChallenge.

[1] https://github.com/seatgeek/fuzzywuzzy